

kinetic-kit

API Reference

v0.1.0

This document lists all public symbols exported by kinetic-kit.

1 Templates

1.1 dissertation

- dissertation()

1.1.1 dissertation

KIT doctoral dissertation template.

- author-title (str | none): Academic title preceding the author's name (e.g. "M.Sc. ").

1.1.1.1 Parameters

```
dissertation(  
  author-title,  
  author-firstname: str,  
  author-surname: str,  
  author-male: bool,  
  title: content,  
  doc-degree: str,  
  doc-degree-f: str,  
  department: str,  
  university-genitive: str,  
  status-approved: bool,  
  exam-date,  
  main-advisor,  
  main-advisor-male: bool,  
  co-advisor,  
  co-advisor-male: bool,  
  format,  
  lang,  
  margin-preset,  
  binding-correction: length,  
  colored-links: bool,  
  draft: bool,  
  draft-info,  
  serif-headings: bool,  
  heading-numbering-depth: int,  
  abstract-en,  
  abstract-de,  
  acknowledgements,  
  notation,  
  abbreviations,  
  show-lof: bool,  
  show-lot: bool,  
  show-lol: bool,  
  own-publications,  
  own-patents,  
  supervised-theses,  
  bibliography,  
  appendix,  
  doc: content  
) -> content
```

1.1.1.1.1 author-firstname str

Author's first name.

Default: "Max"

1.1.1.1.2 author-surname str

Author's surname.

Default: "Mustermann"

1.1.1.1.3 author-male bool

true for male, false for female grammatical forms.

Default: `true`

1.1.1.1.4 title content

Dissertation title.

Default: [Your Thesis Title]

1.1.1.1.5 doc-degree str

Degree name in masculine form.

Default: `"Doktors der Ingenieurwissenschaften (Dr.-Ing.)"`

1.1.1.1.6 doc-degree-f str

Degree name in feminine form.

Default: `"Doktorin der Ingenieurwissenschaften (Dr.-Ing.)"`

1.1.1.1.7 department str

Faculty / department name.

Default: `"KIT-Fakultät für Maschinenbau"`

1.1.1.1.8 university-genitive str

University name in genitive case.

Default: `"des Karlsruher Instituts für Technologie (KIT)"`

1.1.1.1.9 status-approved bool

false = submitted, true = approved.

- exam-date (str | none): Date of oral examination (when approved).
- main-advisor (str | none): Main referee (when approved).

Default: `false`

1.1.1.1.10 main-advisor-male bool

Grammatical gender for main advisor label.

- co-advisor (str | none): Co-referee (when approved).

Default: `true`

1.1.1.1.11 co-advisor-male bool

Grammatical gender for co-advisor label.

- format ("a5" | "17x24" | "a4"): Paper format — "a5" (148×210 mm, default), "17x24" (170×240 mm), or "a4" (210×297 mm, discouraged by KSP). Font sizes and margins are set automatically.
- lang ("de" | "en"): Document language.
- margin-preset ("short" | "medium" | "long"): Margin profile keyed on page count.

Default: **true**

1.1.1.1.12 binding-correction length

BCOR added to inside margin. Default 0mm.

Default: **0mm**

1.1.1.1.13 colored-links bool

KIT Blue links when true, black when false.

Default: **true**

1.1.1.1.14 draft bool

Show "ENTWURF" watermark when true.

- draft-info (str | none): Optional version string below watermark. Default none.

Default: **false**

1.1.1.1.15 serif-headings bool

Use serif font for headings when true. Default false (sans-serif).

Default: **false**

1.1.1.1.16 heading-numbering-depth int

Deepest heading level to number. Default 3.

- abstract-en (content | none): English abstract. none = omit.
- abstract-de (content | none): German abstract. none = omit.
- acknowledgements (content | none): Acknowledgements. none = omit.
- notation (content | none): Notation list. none = omit.
- abbreviations (content | none): Abbreviations list. none = omit.

Default: **3**

1.1.1.1.17 show-lof `bool`

Include List of Figures.

Default: `true`

1.1.1.1.18 show-lot `bool`

Include List of Tables.

Default: `true`

1.1.1.1.19 show-lol `bool`

Include List of Listings.

- own-publications (content | none): Own publications content (heading added by template). none = omit.
- own-patents (content | none): Own patents content (heading added by template). none = omit.
- supervised-theses (content | none): Supervised theses content (heading added by template). none = omit.
- bibliography (content | none): Bibliography content. Pass `bibliography("refs.bib", title: none, style: "ieee")`. The template adds a translated heading. none = omit.
- appendix (content | none): Appendix chapters. Template applies A, A.1, ... numbering and places the appendix at the very end, after all back-matter lists. none = omit.

Default: `false`

1.1.1.1.20 doc `content`

Main document body (chapters only).

1.2 thesis

- thesis()

1.2.1 thesis

KIT Master's / Bachelor's / Diploma thesis template.

1.2.1.1 Parameters

```
thesis(  
  author-firstname: str,  
  author-surname: str,  
  title: content,  
  thesis-type: str,  
  department: str,  
  university-genitive: str,  
  examiner,  
  supervisor,  
  date-submitted,  
  format,  
  lang,  
  margin-preset,  
  binding-correction: length,  
  colored-links: bool,  
  draft: bool,  
  draft-info,  
  serif-headings: bool,  
  heading-numbering-depth: int,  
  abstract-en,  
  abstract-de,  
  acknowledgements,  
  abbreviations,  
  show-lof: bool,  
  show-lot: bool,  
  show-lol: bool,  
  bibliography,  
  appendix,  
  doc: content  
) -> content
```

1.2.1.1.1 author-firstname str

Author's first name.

Default: "Max"

1.2.1.1.2 author-surname str

Author's surname.

Default: "Mustermann"

1.2.1.1.3 title content

Thesis title.

Default: [Your Thesis Title]

1.2.1.1.4 thesis-type `str`

e.g. "Masterarbeit", "Bachelorarbeit".

Default: "Masterarbeit"

1.2.1.1.5 department `str`

Faculty / department name.

Default: "KIT-Fakultät für Maschinenbau"

1.2.1.1.6 university-genitive `str`

University name in genitive case.

- examiner (str | none): First examiner. none if unknown.
- supervisor (str | none): Supervisor. none if unknown.
- date-submitted (str | none): Submission date string. none if unknown.
- format ("a5" | "17x24" | "a4"): Paper format — "a5" (148×210 mm, default), "17x24" (170×240 mm), or "a4" (210×297 mm). Font sizes and margins are set automatically.
- lang ("de" | "en"): Document language.
- margin-preset ("short" | "medium" | "long"): Margin profile keyed on page count.

Default: "des Karlsruher Instituts für Technologie (KIT)"

1.2.1.1.7 binding-correction `length`

BCOR added to inside margin. Default 0mm.

Default: 0mm

1.2.1.1.8 colored-links `bool`

KIT Blue links when true, black when false.

Default: true

1.2.1.1.9 draft `bool`

Show "ENTWURF" watermark when true.

- draft-info (str | none): Optional version string below watermark. Default none.

Default: false

1.2.1.1.10 serif-headings `bool`

Use serif font for headings when true. Default false (sans-serif).

Default: false

1.2.1.1.11 heading-numbering-depth `int`

Deepest heading level to number. Default 3.

- `abstract-en` (content | none): English abstract. none = omit.
- `abstract-de` (content | none): German abstract. none = omit.
- `acknowledgements` (content | none): Acknowledgements. none = omit.
- `abbreviations` (content | none): Abbreviations list. none = omit.

Default: `3`

1.2.1.1.12 show-lof `bool`

Include List of Figures.

Default: `true`

1.2.1.1.13 show-lot `bool`

Include List of Tables.

Default: `true`

1.2.1.1.14 show-lol `bool`

Include List of Listings.

- `bibliography` (content | none): Bibliography content. Pass `bibliography("refs.bib", title: none, style: "ieee")`. The template adds a translated heading. none = omit.
- `appendix` (content | none): Appendix chapters. Template applies A, A.1, ... numbering and places the appendix at the very end, after all back-matter lists. none = omit.

Default: `false`

1.2.1.1.15 doc `content`

Main document body (chapters only).

2 Style Constants

The `kit-style` dictionary exposes the template's visual constants so that custom figures, diagrams, and other content can match the document's typography and KIT color palette exactly.

```
#import "@local/kinetic-kit:0.1.0": kit-style

#let font-sizes = kit-style.font-sizes-by-format.at("a5")
#set text(font: kit-style.fonts.sans, size: font-sizes.small)
#rect(fill: kit-style.colors.green15, stroke: kit-style.colors.green)
```

2.1 Fonts

Field	Value	Description
<code>fonts.serif</code>	<code>("Libertinus Serif",)</code>	Serif body text

Field	Value	Description
fonts.sans	("Libertinus Sans",)	Sans-serif headings and headers
fonts.mono	("Libertinus Mono",)	Code listings
leading	0.75em	Paragraph line spacing ($\approx 1.15\times$ at 10 pt)

2.2 Font Sizes

font-sizes-by-format is a dictionary keyed by format string ("a5", "17x24", "a4"). Each entry contains the sizes below. Access them with `kit-style.font-sizes-by-format.at(format)`.

All fields are length. Example values shown for the "a5" format (10 pt base).

Field	A5 / 17×24	A4	Description
base	10pt	11pt	Body text
chapter	18pt	25pt	Chapter heading (level 1)
section	14pt	17pt	Section heading (level 2)
subsection	12pt	14pt	Subsection heading (level 3)
subsubsection	10pt	11pt	Subsubsection heading (level 4+)
small	8pt	9pt	Running headers, footnotes, captions
footnote	8pt	9pt	Footnote text
title	18pt	25pt	Title page — document title
author	14pt	14pt	Title page — author name
title-info	14pt	14pt	Title page — info lines

2.3 Colors

All values are color. KIT brand colors follow a systematic naming scheme: base name plus optional opacity suffix (70, 50, 30, 15 = 70 %, 50 %, 30 %, 15 % tint).

Field	Hex	Description
colors.green	#009682	KIT Green — primary brand color
colors.green70	#4CB5A7	
colors.green50	#7FCAC0	
colors.green30	#B2DFD9	
colors.green15	#D9EFEC	
colors.blue	#4664AA	KIT Blue — links and accents
colors.blue70	#7D92C3	
colors.blue50	#A2B1D4	
colors.blue30	#C7D0E5	
colors.blue15	#E3E8F2	
colors.black	#000000	
colors.black70	#4D4D4D	
colors.black50	#808080	
colors.black30	#B3B3B3	
colors.black15	#D9D9D9	
colors.palegreen	#82BE3C	KIT extended palette

Field	Hex	Description
colors.yellow	#FAE614	
colors.orange	#DCA01E	
colors.brown	#A08232	
colors.red	#A01E28	
colors.lilac	#A00078	
colors.cyanblue	#50AAE6	
colors.keyword	#0000C8	Syntax highlighting — keyword
colors.comment	#3F7F5F	Syntax highlighting — comment
colors.string	#700055	Syntax highlighting — string literal

3 Components

The components module exports the individual building blocks for assembling a document without the full `dissertation()` / `thesis()` orchestrator. All symbols are accessed through the `components` namespace:

```
#import "@local/kinetic-kit:0.1.0": components, kit-style

#let format = "a5"
#let font-sizes = kit-style.font-sizes-by-format.at(format)

#show: components.setup-page.with(format: format, margin-preset: "short", lang: "de")
#show: components.setup-front-matter
#components.print-toc(font-sizes, lang: "de")
```

You are responsible for applying the wrappers in the correct order: `setup-page` → `setup-front-matter` → `content` → `setup-content` → (optionally) `setup-appendix`.

3.1 Page Setup

3.2 page-setup

- `setup-page()`
- `setup-front-matter()`
- `setup-content()`
- `setup-appendix()`

3.2.1 setup-page

Apply the full KIT document style: page geometry, running headers, KSP typography, heading styles, figure captions, equations, and code blocks. Use as a show rule: `#show: setup-page.with(...)`.

3.2.1.1 Parameters

```
setup-page(  
  format: str,  
  margin-preset: str,  
  lang: str,  
  binding-correction: length,  
  colored-links: bool,  
  draft: bool,  
  draft-info: content,  
  serif-headings: bool,  
  heading-numbering-depth: int,  
  doc: content  
) -> content
```

3.2.1.1.1 format str

Paper format — "a5" (148×210 mm, default), "17x24" (170×240 mm), or "a4" (210×297 mm).
Font sizes and margins are set automatically per KSP specifications.

Default: "a5"

3.2.1.1.2 margin-preset str

Margin profile keyed on expected page count — "short" (under 200 pp), "medium" (200–399 pp),
"long" (400+ pp).

Default: "short"

3.2.1.1.3 lang str

Document language — "de" or "en".

Default: "de"

3.2.1.1.4 binding-correction length

Extra inside margin added for binding (e.g. 3mm).

Default: 0mm

3.2.1.1.5 colored-links bool

Render external hyperlinks in KIT blue when true.

Default: true

3.2.1.1.6 draft bool

Show the draft watermark on every page when true.

Default: false

3.2.1.1.7 draft-info `content`

Optional extra text appended to the watermark (e.g. a git SHA).

Default: `none`

3.2.1.1.8 serif-headings `bool`

Use Libertinus Serif for headings when true. Default false

Default: `false`

3.2.1.1.9 heading-numbering-depth `int`

Deepest heading level that receives a number. Default 3. Headings deeper than this are styled normally but rendered without a number or indent grid.

Default: `3`

3.2.1.1.10 doc `content`

Document body (injected automatically by the show rule).

3.2.2 setup-front-matter

Switch to Roman numeral page numbering and remove heading numbering. Apply before front-matter content: `#show: setup-front-matter.`

3.2.2.1 Parameters

`setup-front-matter`(doc: `content`) -> `content`

3.2.2.1.1 doc `content`

Document body (injected automatically by the show rule).

3.2.3 setup-content

Switch to Arabic page numbering and enable 1.1 heading numbering. Apply before the main content: `#show: setup-content.`

3.2.3.1 Parameters

`setup-content`(doc: `content`) -> `content`

3.2.3.1.1 doc `content`

Document body (injected automatically by the show rule).

3.2.4 setup-appendix

Switch to A.1 heading numbering and reset the heading counter. Apply before appendix chapters:
#show: setup-appendix.

3.2.4.1 Parameters

```
setup-appendix(doc: content) -> content
```

3.2.4.1.1 doc content

Document body (injected automatically by the show rule).

3.3 Title Pages

3.4 title-page

- print-dissertation-title()
- print-thesis-title()

3.4.1 print-dissertation-title

Render the KIT dissertation title page (German legal format). All text on this page is always in German regardless of document language.

3.4.1.1 Parameters

```
print-dissertation-title(  
  title: content,  
  author-title: content,  
  author-firstname: str,  
  author-surname: str,  
  author-male: bool,  
  doc-degree: str,  
  doc-degree-f: str,  
  department: str,  
  university-genitive: str,  
  status-approved: bool,  
  exam-date: content,  
  main-advisor: content,  
  main-advisor-male: bool,  
  co-advisor: content,  
  co-advisor-male: bool,  
  format: str  
) -> content
```

3.4.1.1.1 title content

Dissertation title.

3.4.1.1.2 author-title content

Academic title preceding the author's name (none = omit).

Default: "M.Sc. "

3.4.1.1.3 author-firstname `str`

Author's first name.

Default: "Max"

3.4.1.1.4 author-surname `str`

Author's surname.

Default: "Mustermann"

3.4.1.1.5 author-male `bool`

Selects grammatical gender for the degree article.

Default: `true`

3.4.1.1.6 doc-degree `str`

Degree name in masculine form (e.g. "Doktor-Ingenieur").

Default: "Doktors der Ingenieurwissenschaften (Dr.-Ing.)"

3.4.1.1.7 doc-degree-f `str`

Degree name in feminine form (e.g. "Doktor-Ingenieurin").

Default: "Doktorin der Ingenieurwissenschaften (Dr.-Ing.)"

3.4.1.1.8 department `str`

KIT department or faculty.

Default: "KIT-Fakultät für Maschinenbau"

3.4.1.1.9 university-genitive `str`

University name in genitive (e.g. "des Karlsruher Instituts...").

Default: "des Karlsruher Instituts für Technologie (KIT)"

3.4.1.1.10 status-approved `bool`

`false` = submitted version, `true` = approved version.

Default: `false`

3.4.1.1.11 exam-date content

Date of oral examination — shown only when status-approved is true.

Default: none

3.4.1.1.12 main-advisor content

Main referee — shown only when status-approved is true.

Default: none

3.4.1.1.13 main-advisor-male bool

Selects gendered label for the main advisor.

Default: true

3.4.1.1.14 co-advisor content

Co-referee — shown only when status-approved is true.

Default: none

3.4.1.1.15 co-advisor-male bool

Selects gendered label for the co-advisor.

Default: true

3.4.1.1.16 format str

Paper format — "a5", "17x24", or "a4". Determines font sizes and title-page margins.

Default: "a5"

3.4.2 print-thesis-title

Render the KIT master's / bachelor's thesis title page (German legal format). All text on this page is always in German regardless of document language.

3.4.2.1 Parameters

```
print-thesis-title(  
  title: content,  
  thesis-type: str,  
  author-firstname: str,  
  author-surname: str,  
  department: str,  
  university-genitive: str,  
  examiner: content,  
  supervisor: content,  
  date-submitted: content,  
  format: str  
) -> content
```

3.4.2.1.1 title content

Thesis title.

3.4.2.1.2 thesis-type str

Thesis type (e.g. "Masterarbeit", "Bachelorarbeit").

Default: "Masterarbeit"

3.4.2.1.3 author-firstname str

Author's first name.

Default: "Max"

3.4.2.1.4 author-surname str

Author's surname.

Default: "Mustermann"

3.4.2.1.5 department str

KIT department or faculty.

Default: "KIT-Fakultät für Maschinenbau"

3.4.2.1.6 university-genitive str

University name in genitive (e.g. "des Karlsruher Instituts...").

Default: "des Karlsruher Instituts für Technologie (KIT)"

3.4.2.1.7 examiner content

First examiner (Erstprüfer).

Default: none

3.4.2.1.8 supervisor content

Supervisor (Betreuer).

Default: none

3.4.2.1.9 date-submitted content

Submission date string.

Default: none

3.4.2.1.10 format str

Paper format — "a5", "17x24", or "a4". Determines font sizes and title-page margins.

Default: "a5"

3.5 Outlines

3.6 outlines

- `setup-outlines()`
- `print-toc()`
- `print-lof()`
- `print-lot()`
- `print-lol()`

3.6.1 setup-outlines

Shared outline styling, applied as a show rule so every outline in the document — including a user's own — matches.

3.6.1.1 Parameters

`setup-outlines`(body: content) -> content

3.6.1.1.1 body content

Document body (injected automatically by the show rule).

3.6.2 print-toc

Print the table of contents, including a separate appendix outline when present.

3.6.2.1 Parameters

```
print-toc(  
    font-sizes: dict,  
    lang: str,  
    serif-headings: bool  
) -> content
```

3.6.2.1.1 font-sizes dict

Format-specific font sizes resolved by the template.

3.6.2.1.2 lang str

Document language — "de" or "en".

Default: "de"

3.6.2.1.3 serif-headings bool

Use serif font for the appendix section title when true.

Default: false

3.6.3 print-lof

Print the list of figures.

3.6.3.1 Parameters

```
print-lof(lang: str) -> content
```

3.6.3.1.1 lang str

Document language — "de" or "en".

Default: "de"

3.6.4 print-lot

Print the list of tables.

3.6.4.1 Parameters

```
print-lot(lang: str) -> content
```

3.6.4.1.1 lang str

Document language — "de" or "en".

Default: "de"

3.6.5 print-lol

Print the list of listings.

3.6.5.1 Parameters

```
print-lol(lang: str) -> content
```

3.6.5.1.1 lang str

Document language — "de" or "en".

Default: "de"

3.7 Figures

3.8 figures

- flex-caption()
- setup-figures()

3.8.1 flex-caption

Two-part caption: short version for LoF/LoT, long version under the figure.

Usage: #figure(..., caption: flex-caption(short: [Short], long: [Long.]))

3.8.1.1 Parameters

```
flex-caption(  
    short: content,  
    long: content  
) -> content
```

3.8.1.1.1 short content

Short caption shown in List of Figures / Tables.

Default: none

3.8.1.1.2 long content

Full caption shown below the figure in the document body.

Default: none

3.8.2 setup-figures

Shared figure, caption, and table styling. Apply as a show rule.

3.8.2.1 Parameters

```
setup-figures(  
    font-sizes: dict,  
    lang: str,  
    body: content  
) -> content
```

3.8.2.1.1 font-sizes dict

Format-specific font sizes resolved by the template.

3.8.2.1.2 lang str

Document language — "de" or "en".

Default: "de"

3.8.2.1.3 body content

Document body (injected automatically by the show rule).

3.9 Headings

3.10 headings

- setup-headings()

3.10.1 setup-headings

Shared heading styling: per-level sizes and spacing, chapter page breaks and counter resets, and number-body alignment. Apply as a show rule.

3.10.1.1 Parameters

```
setup-headings(  
  font-sizes: dict,  
  serif-headings: bool,  
  heading-numbering-depth: int,  
  body: content  
) -> content
```

3.10.1.1.1 font-sizes dict

Format-specific font sizes resolved by the template.

3.10.1.1.2 serif-headings bool

Use Libertinus Serif for headings when true.

Default: false

3.10.1.1.3 heading-numbering-depth int

Deepest heading level that receives a number.

Default: 3

3.10.1.1.4 body content

Document body (injected automatically by the show rule).